Zidu Zhang

PhD Applicant in Computer Vision, Multimodal AI, and Intelligent Systems

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Education

Beijing University of Technology Beijing, China

B.Eng. in Computer Science and Technology, expected Sep. 2023 - Jun. 2027

- GPA: 3.2/4.0.
- Research interests: computer vision, multimodal learning, graph/hypergraph modeling, time-series forecasting, and LLM agents.

Research Experience

Summer Research Intern Jun. 2026 - Aug. 2026

University of Virginia *United States*

- Built a state-level influenza early-warning data loop for CDC FluView/ILINet, lab positivity, ILI Activity Map, CDC text reports, Google Trends, NASA/NOAA weather, hospitalization, wastewater, and holiday signals, aligned into `state-week` panels for 1-8 week `%ILI` forecasting and onset/surge warning.
- Designed ReLiEF-HGDiff v3.1 with release-aware hypergraphs, AS-3D RoPE, reliability learning, trend filtering, prototype libraries, regime-conditioned quantile decoding, and spatial tree calibration to detect early abnormal signals under reporting lag.
- Established leakage-controlled backtesting against persistence, seasonal naive, Ridge, DLinear/Patch, ARGO-like, Cola-like, and multimodal ExtraTrees baselines; achieved 0.6477 `%ILI` MAE and improved surge recall from 0.4477 to 0.5891 with the warning-enhanced model.

Atlas Repository-Level LLM Integration Study 2026

ESEM 2026 Submission *Research Project*

- Designed and implemented Atlas, a repository-level analysis framework for identifying real LLM component integration across source code, dependency files, configuration files, READMEs, Dockerfiles, issues, and commits.
- Built LLM keyword libraries and component-normalization dictionaries for model families, inference frameworks, API providers, deployment backends, and parameters; combined multi-source evidence fusion with LLM-assisted semantic reasoning for role, loading/deployment, provider, and usage classification.
- Created a 200-repository manually annotated benchmark and applied Atlas to 12,991 active GitHub repositories, identifying 1,036 LLM-integrated projects, 75 plaintext API-credential leaks, and 70 license-compliance risks; achieved 89.11% repository-identification F1 and 84.97% loading-pattern F1.

Research Assistant Jul. 2025 - Jul. 2026

North China Energy Internet Research Institute *China*

- Developed a multi-source time-series pipeline for agency power-purchase load forecasting with pandas and NumPy, covering anomaly detection, missing-value repair, temporal alignment, and 100,000+ trainable samples for multimodal forecasting.
- Conducted field research on power-purchase workflows, load formation mechanisms, and data-collection chains; translated operational constraints into feature engineering designs that improved business fit.
- Integrated LLM agents and SHAP-based analysis for event extraction, abnormal-load attribution, and forecast correction, reducing representative forecasting error by approximately 15%-64% and improving abnormal-period stability by approximately 13%.

Power Metering Reliability and Intelligent Operations with Large Models Selected project

- Built an LLM- and knowledge-graph-based fault diagnosis workflow for power metering equipment; collected 40,000+ device, vendor, and fault records with Selenium from industrial web sources.
- Implemented YOLOv8 + OCR meter-panel recognition and collaborated on Flask, Jinja2, and Neo4j components for PowerKG entity/relation ingestion and MeterAgent diagnosis; reached 97%+ panel-detection accuracy and improved effective OCR recognition by about 15%.

Multimodal Load Forecasting and Intelligent Analysis for Agency Power Purchase Selected project

- Led the transaction-curve forecasting module using PyTorch, Transformer, ViT, BERT, LLM agents, and SHAP to combine load, weather, text, and event signals for multimodal load prediction.
- Built the data-processing pipeline, model architecture, training/tuning workflow, and intelligent analysis integration; reduced representative forecasting error by approximately 15%-64% and improved abnormal-period stability by approximately 13%.

HGDefect-YOLO: Industrial Surface Small Defect Detection Selected project

- Developed a YOLO-based detector for small, low-contrast, and irregular industrial surface defects by redesigning the backbone and neck while keeping the detection head unchanged.
- Proposed PPAP-EMA, a KDE-guided Hypergraph Neck, and RGCU-CLAG for peak-preserving attention, high-order cross-scale interaction, and edge-texture recovery.

- Achieved 79.9% AP50 and 0.459 AP50:95 on NEU-DET with 5.62M parameters and 14.85 GFLOPs; improved crazing AP to 65.8%, outperforming YOLOv8 by +22.1 pp and RAGA-YOLO by +30.2 pp.

GeoMask-OMR: Piano Score Recognition and MusicXML Reconstruction Selected project

- Designed an OMR pipeline for complex piano-score images covering staff localization, symbol detection, mask refinement, relation modeling, note/rest assembly, MusicXML export, and automatic evaluation.
- Combined OpenCV/NumPy baselines with DEIM-D-FINE, SAM2, DINOv2, and RapidOCR; converted rule-based outputs into COCO weak labels and built scripts for training, inference, visualization, and CER/SER/LER evaluation.
- Built an 81-class DeepScores symbol dataset with 1,362 training images, 352 validation images, about 872,000 training annotations, and 240,000 validation annotations; generated 6,485 fused symbols and 2,187 notes on five real score pages.

SCH-MDETR: Micro UAV Detection in Visible Images Selected project

- Proposed SCH-MDETR, a scale-adaptive CLIP-guided hypergraph Mamba-DETR framework for visible micro-UAV detection under weak texture, small object scale, and cluttered-background conditions.
- Designed a Dynamic Top-K Scale Router and Unified Hypergraph Reasoning module with CLIP semantic prototypes to fuse scale-aware visual features, high-order object relations, and semantic priors for more stable localization.
- Achieved 87.37% mAP50 and 51.60% mAP50-95 on Det-Fly validation, improving RT-DETR-R18-N mAP50-95 by 1.72 percentage points.

AI-Driven Dynamic Tiering Platform for Smart Charging Stations Selected project

- Designed the demand forecasting, collaborative operation, dynamic tiering, and platform-visualization workflow for charging-station management using order, location, traffic, weather, holiday, functional-zone, and grid-load data.
- Proposed a platform plan combining dynamic pricing, ordered charging guidance, multi-agent dispatch, warning work orders, and user navigation for intelligent station operations.

Full-Chain Intelligent Police System for Vehicle Violation Governance Selected project

- Designed an edge-cloud anomaly perception workflow for flipped plates, unlicensed vehicles, cloned plates, and modified trucks using YOLO, VLM, CRNN, ReID, and CLIP.
- Integrated checkpoint trajectories, spatiotemporal relationship graphs, and clustering analysis to support the chain from anomaly discovery to identity recovery and trajectory enforcement.

Publications and Manuscripts

[1] **Zidu Zhang**, Ruidi Liu, and Xinyi Chen. "YOLO-ARC: Deployment-Ready and Robust Small-Defect Detection for Industrial Surfaces." *ICPECA 2026*. Published, EI-indexed conference.

[2] **Zidu Zhang**, et al. "RMPF-Orion: Proxy Electricity Load Forecasting with Robust Temporal Modeling and Multimodal Learning." Submitted to *IEEE Transactions on Smart Grid (TSG)*. First author.

[3] **Zidu Zhang**, Mengfei Xiao, Zelin Li, Jialiang Dong, Zhitao Guan, Willy Susilo, and Siqi Ma. "Echoes of Change: A Large-Scale Empirical Study of LLM Integration in Open-Source Software." Submitted to *ESEM 2026*. CCF-B; CORE A.

[4] **Zidu Zhang**, Rui Jiang, Ximai Duan, Zihan Gao, Mengfei Xiao, and Xuan Li. "HGDefect-YOLO: Peak-Preserving High-Order Feature Learning for Industrial Surface Defect Detection." Submitted to *BMVC 2026*. CCF-C; CORE A.

[5] **Zidu Zhang**, et al. "SCH-MDETR: Scale-Adaptive CLIP-Guided Hypergraph Mamba-DETR for UAV Detection." Submitted to *ACCV 2026*. CCF-C; CORE B.

[6] **Zidu Zhang**, et al. "ReLiEF: Release-Aware Hypergraph Learning for As-Of Multimodal Influenza Early Warning." Planned submission to *AAAI 2027*. CCF-A; CORE A*.

[7] **Zidu Zhang**, et al. "GeoMask-OMR: Staff-Aware Neural Symbol Recognition and Structural Reconstruction for Complex Piano Score Images." Planned submission to *AAAI 2027*. CCF-A; CORE A*.

Honors and Awards

- National Second Prize, 12th National Undergraduate Energy Economics Academic Creativity Competition, 2025.
- National Second Prize, RAICOM Robotics Developer Competition.
- Beijing Special Prize, "Youth Innovation Beijing" 2026 Challenge Cup Capital University Student Entrepreneurship Plan Competition.
- Beijing Second Prize, Higher Education Press Cup National Undergraduate Mathematical Contest in Modeling, 2025.

Skills

Algorithms and models: YOLO, classification/segmentation, OCR, multimodal learning, Transformer, graph/hypergraph modeling, time-series forecasting, reinforcement learning, and PyTorch training/evaluation.
LLMs and agents: SFT/RLHF, prompt engineering, RAG, multi-agent systems, event extraction, result optimization, ViT, Stable Diffusion, and MLLMs.
Engineering: Python, C/C++, R, MATLAB, Linux, Git, Docker, Pandas/NumPy, Flask/Jinja2, Neo4j, and distributed training.